Seongmuk Khang

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Seoul, South Korea

Research Interest

My research focuses on quantum algorithms and quantum-classical hybrid methods, with an emphasis on identifying the conditions under which practical quantum advantage can be achieved. I am interested in developing theoretically grounded, hardware-aware algorithms that exploit problem-specific structure and remain effective under realistic resource constraints.

Education

Chiba University *Apr 2018 – Mar 2020*
M.Eng. in Applied and Cognitive Informatics, Division of Mathematics and Informatics Chiba, Japan
- Thesis: *Quantum Algorithms for Finding All Eigenvalues of Sparse Hermitian Matrices*
- Advisor: Prof. Yuichiro Fujiwara
- GPA (WES evaluated): 3.94/4.00

Chiba University *Apr 2014 – Mar 2018*
B.Eng. in Informatics and Imaging Systems, Faculty of Engineering Chiba, Japan
- Thesis: *The Lower Bound on the Maximum Generalized Independent Number*
- Advisor: Prof. Yuichiro Fujiwara
- GPA (WES evaluated): 3.60/4.00

Research Experience

Qunova Computing [🌐] *May 2026 – Present*
Quantum Algorithm Researcher Daejeon and Seoul, South Korea

Project: Hybrid Quantum–Classical Optimization for the Shortest Vector Problem (Confidential)

- Develop hybrid quantum–classical optimization algorithms for the shortest vector problem (SVP), a core hard problem in lattice-based cryptography, with a focus on scalability under realistic hardware constraints

Independent Research *May 2025 – Present*
Seoul, South Korea

Project: *Local Scrambling and Structural Gradient Bias in Hardware-Efficient Quantum Circuits*
(Under revision at *Quantum Information Processing*; see [S.1])

- Analyzed trainability of hardware-efficient quantum circuits by simulating gradient behavior across circuit depths, cost placements, and gate layouts
- Quantified structural gradient bias through segment-wise gradient energy measurements, identifying parameter regions with dominant or suppressed gradient contributions
- Tested local scrambling strategies to improve gradient propagation while examining trade-offs in convergence behavior and cost-function dependence

Project: Extension of *Quantum Algorithms for Finding All Eigenvalues of Sparse Hermitian Matrices*

- Showed that resolving all N eigenvalues requires QPE precision at the scale of the minimum eigenvalue spacing $\Delta_{\min} = O(1/N)$; under Heisenberg-limited scaling, the total cost becomes $\tilde{O}(N)$ samples $\times$ $\Omega(1/\Delta_{\min})$ evolution time per sample
- Concluded that the speedup is at most polynomial: marginal against sparse classical methods when $\Delta_{\min} = \Theta(1/N)$, and lost for more closely spaced spectra

Chiba University *May 2019 – Mar 2020*
Graduate Researcher Chiba, Japan

Project: *Quantum Algorithms for Finding All Eigenvalues of Sparse Hermitian Matrices*

- Developed a quantum algorithm that reconstructs the full spectrum of a sparse Hermitian matrix by combining quantum phase estimation (QPE) with coupon-collector sampling
- Removed the need for eigenstate preparation by running QPE on random computational-basis states, which sample each eigenvalue with average probability $1/N$
- Derived an expected complexity of $\tilde{O}(Ns^2 \log^2 N)$ at fixed precision for s-sparse matrices

Patents and Publications

P=Patent, S=In Submission

[S.1] **Seongmuk Khang**. (2026). ***Local scrambling and structural gradient bias in hardware-efficient quantum circuits***. Under revision at *Quantum Information Processing*. Preprint: Research Square, DOI: 10.21203/rs.3.rs-8232783/v1.

[P.1] **Seongmuk Khang,** et al. (2022). **Method and Device for Predicting Patient Prognosis**. Korean Patent Application No. 10-2022-0069591. Filed Jun. 8, 2022; published as KR 10-2023-0069798 A, May 19, 2023. Korean Intellectual Property Office.

Professional Experience

Acryl Inc. [🌐] *Sep 2020 – Mar 2025*
Senior Research Engineer Seoul, South Korea

- **Medical AI:** Developed AI models for COVID-19 screening, COVID-19 prognosis, and pressure-ulcer detection in partnership with Samsung Medical Center
- **COVID-19 Prognosis:** Adapted model architecture and loss functions based on clinical-data analysis; outperformed two other participating companies by ∼0.05 in AUROC, leading to a patent application [P.1]
- **Open-Source LLM:** Led the data team; introduced meaning-preserving paraphrase augmentation and a new training task for non-question data, contributing to a 1st-place ranking (open-source category) on the Horangi LLM Leaderboard for four consecutive months

Teaching and Mentoring Experience

Chiba University *Fall 2018, Fall 2019*
Instructor, Basic Mathematics

- Taught foundational mathematics to international students in the Japan-Korea Joint Program for Science and Engineering, supporting their transition to university-level quantitative coursework

Chiba University *Apr 2018 – Mar 2019*
Academic Mentor, Japan-Korea Joint Program for Science and Engineering

- Mentored first-year international undergraduates on mathematics coursework, study strategies, and adaptation to university life in Japan

Kkumnamu Community Child Center *Oct 2012 – Jan 2013*
Volunteer Tutor

- Tutored middle school students in academic subjects through 63 hours of volunteer service across 22 sessions

Honors and Awards

Horangi LLM Leaderboard: Open-Source Category, 1st Place *Jul 2024 – Oct 2024*
Weights & Biases (W&B) [🌐]

- Ranked 1st among open-source LLMs for four consecutive months, as a member of Acryl's LLM team

Japan-Korea Joint Scholarship Program for Science and Engineering Students *Mar 2013 – Mar 2018*
Governments of Korea and Japan [🌐]

- Merit-based government scholarship covering full tuition, monthly stipends, and a preparatory year of Japanese language and academic training

Skills

- **Technical Skills:** Python, Qiskit, PennyLane, PyTorch, NumPy, SciPy, pandas, scikit-learn, Git, Docker
- **Languages:** Korean: Native; English: Professional (TOEFL iBT 102/120; R: 29, L: 23, S: 24, W: 26); Japanese: Fluent

Leadership Experience

Chiba University: Traditional Music Club *Jan 2016 – Mar 2017*
President

- Led a student club promoting Korean traditional music; facilitated Samulnori instruction at a UNESCO-associated school and performed at multicultural events, including the Yokohama Multicultural Parade